

$$y = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \dots + \theta_k x_k$$

Suppose we change x_j slightly by Δx_j & keep all other values as it is.
then notice the change in y . (Δy)

$$\Delta y = \theta_j \Delta x_j$$

$$\boxed{\frac{\Delta y}{\Delta x_j} = \theta_j} \quad (j=1, 2, \dots, k)$$

θ_j high value means a small change in x_j can have significant impact on target variable.

small θ_j means the change in the response variable will not be significant for significant change in x_j

$$y = \theta_0 + \theta_1 x \rightarrow (x + \Delta x)$$

$$\Rightarrow y + \Delta y = \theta_0 + \theta_1 \cdot (x + \Delta x)$$

$$\Rightarrow y + \Delta y = \underbrace{\theta_0 + \theta_1 x}_y + \theta_1 \Delta x$$

$$\Rightarrow \Delta y = \theta_1 (\Delta x)$$